CSCI 5521: Machine Learning Fundamentals (Spring 2022)

Quiz 3 (Thurs, Mar 31))

Due on Gradescope at 02:00 PM, Friday, Apr 1

Instructions:

- This quiz has 3 questions, 30 points, on 1 page.
- Please **write your name & ID on this cover page**.

1. (**9 points**) Select all the correct statement(s).
   (a) A Perceptron (without activation functions) is a linear model.
   (b) A Multilayer Perceptron (without activation functions) is a linear model.
   (c) A Perceptron can be used for both classification and regression.

2. (**13 points**) Pick one parameter in a (Multilayer) Perceptron, and write its update equation. The parameter can be for a node at any layer and for either classification or regression. Start with writing the objective function.

3. (**8 points**) Give an example of an activation function in Multilayer Perceptron. You can write either the name or the mathematical representation of the function. Briefly discuss roles of activation functions in a Multilayer Perceptron.